Comparing human and LLM politeness strategies in free production

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Abstract

Polite speech poses a fundamental alignment challenge for large language models (LLMs). Humans deploy a rich repertoire of linguistic strategies to balance informational and social goals – from positive approaches that build rapport (compliments, expressions of interest) to negative strategies that minimize imposition (hedging, indirectness). We investigate whether LLMs employ a similarly context-sensitive repertoire by comparing human and LLM responses to English-language scenarios in both constrained and open-ended production tasks. We find that larger models ($\geq$70B parameters) successfully replicate key effects from the computational pragmatics literature, and human evaluators prefer LLM-generated responses in open-ended contexts. However, further linguistic analyses reveal that models disproportionately rely on negative politeness strategies to create distance even in positive contexts, potentially leading to misinterpretations. While LLMs thus demonstrate an impressive command of politeness strategies, these systematic differences provide important groundwork for making intentional choices about pragmatic behavior in human-AI communication.

1 Introduction

Speakers do not always say exactly what they mean. For example, we might say a friend's poem "wasn't terrible" rather than saying "it was bad" to avoid hurting their feelings (Yoon et al., 2020), or just compliment specific elements that we liked without mentioning other elements we didn't like (Goffman, 1967; Pinker et al., 2008). These kinds of politeness strategies allow speakers to balance competing goals, conveying accurate information while maintaining positive relationships (Hill et al., 1986; Leech, 2014). As large language models (LLMs) are increasingly deployed in open-ended interactions across sensitive social domains like healthcare and education, their ability to appropriately use and understand polite language remains an important alignment challenge.

Politeness theory provides a valuable framework for addressing these questions. Seminal work by Brown and Levinson (1987) distinguishes between positive politeness strategies that affirm the listener (compliments, expressions of interest) and negative politeness strategies that minimize imposition (hedging, indirectness). While subsequent work has expanded this framework to encompass broader relational and rapport management concerns (Spencer-Oatey, 2011; Watts, 2003; Locher, 2013), this basic distinction remains crucial: different contexts call for different strategies, and mismatches can lead to communication breakdowns. If an AI system employs negative hedging strategies ("I am somewhat concerned that this approach might not be optimal") in contexts where human speakers would expect positive, rapport-building strategies ("I love your creativity here, and wonder if we could build on it by..."), users may be left unsure whether the hedging reflects a genuinely negative evaluation or simply a systematic bias in expressions of politeness.

This kind of pragmatic misalignment represents a critical gap in our understanding of LLMs as social agents. While considerable attention has been paid to whether models can recognize politeness or generate polite language in constrained settings, effective social interaction depends not just on understanding politeness norms in the abstract but on actively selecting and applying appropriate strategies from a diverse linguistic repertoire. Human speakers navigate this complexity intuitively, deploying hedging, elaboration, indirect speech acts, and numerous other strategies to balance competing communicative goals in context-sensitive ways. To fully understand LLMs' grasp of politeness strategies, we need to examine whether they exhibit similar patterns of strategy selection and deployment across different contexts. This requires

moving beyond limited-choice evaluations to examine open-ended language generation, where models have access to the full range of linguistic choices. Rather than prescribing how AI systems ought to handle politeness, we seek descriptive evidence about current patterns of behavior.

In this work, we investigate we compare human and LLM politeness strategies in English-language scenarios, making the following contributions:

- We test whether LLMs reproduce human patterns of goal sensitivity in polite feedback using constrained response sets from Yoon et al. (2020).
- We collect and analyze a new dataset of open-ended responses from both humans and LLMs to identical social scenarios, enabling direct comparison of politeness strategies.
- We perform detailed linguistic analyses to identify systematic differences in how humans and LLMs deploy various categories of politeness strategies.

Our results reveal that while LLMs have acquired important aspects of human-like pragmatic competence in polite language production – enough to be preferred by human evaluators – they also show systematic differences in strategy deployment that raise intriguing questions about the mechanisms underlying their social language capabilities. In particular, we find that models disproportionately rely on negative politeness strategies (minimizing imposition) even in contexts where humans prefer positive politeness strategies (building rapport), suggesting important differences in how these systems navigate social interactions[1].

2 Related Work

2.1 Computational models of politeness

Research on politeness in linguistics and cognitive science has evolved from descriptive frameworks to quantitative models of pragmatic language use. Foundational work by Brown and Levinson (1987) established a systematic taxonomy of strategies, which has provided conceptual scaffolding for subsequent computational approaches. Studies in computational linguistics have since documented various linguistic markers of politeness across languages and contexts, examining formal features such as hedging, indirectness, and specific syntactic constructions correlate with perceived politeness (Danescu-Niculescu-Mizil et al., 2013; Aubakirova and Bansal, 2016).

Recent models in the Rational Speech Act (RSA) framework have explained the use of polite language as emerging from tradeoffs between informational utility capturing the desire for accuracy, a social utility representing the goal of making listeners feel good, and a self-presentational term reflecting speakers' desire to be perceived as both kind and honest (Yoon et al., 2020; Lumer and Buschmeier, 2022; Carcassi and Franke, 2023; Gotzner and Scontras, 2024). This body of work has established a solid theoretical foundation for analyzing politeness as a pragmatic phenomenon arising from underlying tradeoffs. However, existing models have primarily focused on explaining choices among a small number of constrained utterance alternatives rather than modeling the rich variety of strategies humans employ in open-ended generation contexts.

2.2 Pragmatic capabilities in LLMs

Recent research has explored various aspects of pragmatic competence in large language models. Studies have examined LLMs' ability to understand indirect speech acts (Ruis et al., 2024; Jian and Narayanaswamy, 2024), recognize conversational implicatures (Hu et al., 2022; Lipkin et al., 2023), and interpret non-literal language (Yerukola et al., 2024; Liu et al., 2024). These investigations predominantly employ multiple-choice formats, presenting models with pragmatic puzzles and evaluating their ability to select contextually appropriate interpretations. Results generally suggest that modern LLMs demonstrate sophisticated pragmatic understanding, often approaching human-like performance on benchmark tasks. However, these studies primarily assess recognition rather than production capabilities, leaving open questions about whether models can actively deploy pragmatic strategies in their own generated outputs.

2.3 Polite language generation in LLMs

Work on generating polite language in AI systems represents a small but growing research area. Early approaches focused on style transfer, with systems like those developed by Niu and Bansal (2018) demonstrating that neural models could transform neutral text into more polite versions through specific syntactic transformations. Subsequent work explored paraphrasing to increase po-

[1]Data & Code: https://github.com/haoranzhao419/politeness-speech-production

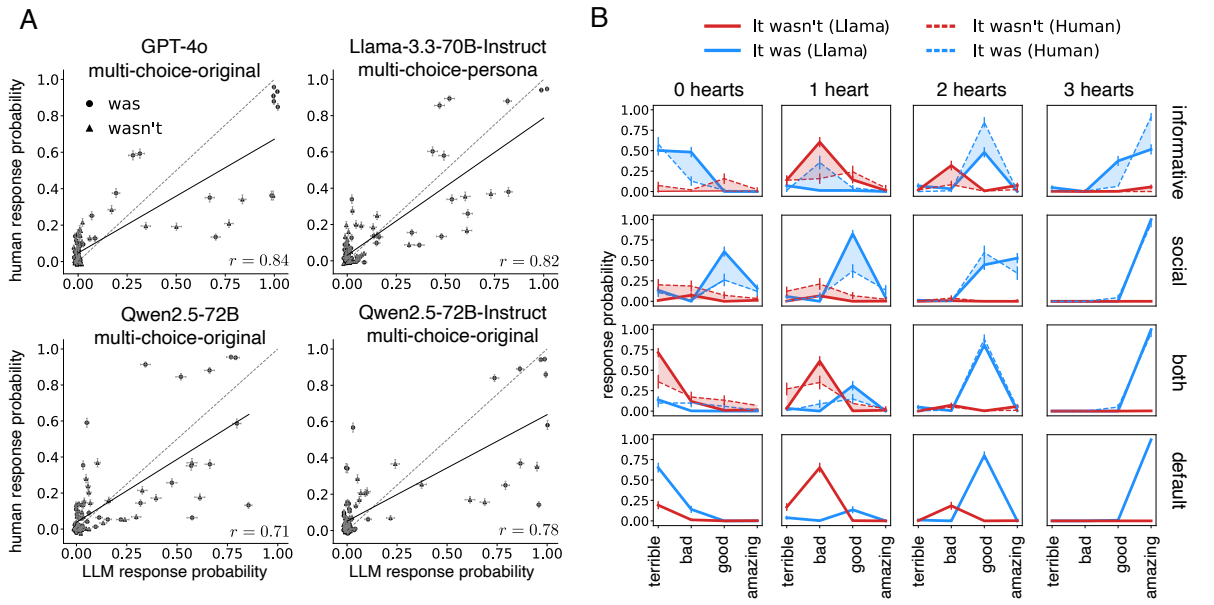

Figure 1: (A) Correlations between human and model response probabilities for the top 4 models with specific prompting strategies we tested. Both the base and instruct-tuned versions of `Qwen2.5-72B` are shown here for comparison. Error bars are 95% confidence intervals across vignettes. (B) Comparing the pattern of human and LLM responses across different communicative goals and ratings. Model results are from `Llama-3.3-70B-Instruct` using the multi-choice-persona prompting strategy; human responses are from Yoon et al. (2020).

liteness (Fu et al., 2020), politeness-focused style transfer (Madaan et al., 2020), and creating polite chatbots (Mukherjee et al., 2023). However, these systems typically focused on surface-level transformations rather than strategic deployment of politeness based on contextual factors. As noted in a recent survey (Priya et al., 2024), existing approaches to polite language generation have predominantly emphasized isolated features (hedging expressions, please markers, specific lexical choices) rather than examining the full repertoire of politeness strategies and how they're selected based on communicative context. This leaves a significant gap in our understanding of whether LLMs can approximate the context-sensitivity that characterizes human politeness. Our work addresses this gap by directly comparing politeness strategies in humans and LLMs across varying communicative goals, examining whether models align with human preferences for positive versus negative politeness strategies in different contexts.

3 Experiment 1: Constrained Settings

To what extent are LLMs sensitive to the goals that give rise to politeness in human speech? To address this question, we first examined whether LLMs could reproduce the patterns of goal-sensitive language use reported by Yoon et al. (2020). Their study provided empirical evidence for a computational model of politeness where speakers strategically balance informational accuracy with social goals. Most notably, they found that when giving negative feedback, humans often deploy indirectness through negation (e.g., "wasn't terrible" rather than "bad"; see also Gotzner and Scontras, 2024; Lumer and Buschmeier, 2022).

We reimplemented this experiment with LLMs to assess their pragmatic competence in a constrained setting. In each scenario, a character gives feedback about another character's performance (e.g., a piano play or presentation), with the true quality ranging from 0 to 3 hearts. The speaker has one of four communicative goals: to be *informative*, to be *kind*, or *both*. We also added a *default* condition with no explicit goal specified to understand how LLMs behave by default. Models selected from the same set of eight responses used by humans, combining either "was" or "wasn't" with four adjectives (terrible, bad, good, amazing).

We tested a range of open-source (8B-72B parameters) and closed-source models using two prompting strategies: an "original" strategy that presented scenarios verbatim, and a "persona" variant that systematically varied speaker characteristics (e.g., gender, occupation, background) to better